

Nifty100 Financial Intelligence Platform

Analyst Guide

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Internship Project

Project Overview

The platform ingests Nifty100 financial data, performs ETL, stores data in SQLite, computes financial ratios, exposes REST APIs using FastAPI, and provides an interactive Streamlit dashboard.

Architecture

Excel Files → ETL → SQLite → Analytics → FastAPI → Streamlit Dashboard → PDF Reports

Database

Tables: companies, profitandloss, balancesheet, cashflow, financial_ratios, market_cap, peer_groups, sectors, stock_prices, analysis, documents, prosandcons, ranking_engine, screening_engine.

ETL Pipeline

Load, clean, normalize, validate, transform, and persist data into SQLite.

Dashboard Guide

Home, Screener, Company Profile, Peer Comparison, Financial Ratios, Charts.

API Guide

Example:

GET /api/v1/health

GET /api/v1/screener

Use curl or Swagger UI.

PDF Reports

Generate company tearsheets summarizing metrics, ratios, charts, and pros/cons.

Troubleshooting

Check database path, dependencies, FastAPI server, Streamlit startup, and API connectivity.

Performance

SQLite indexes added on company_id and year. API load testing and dashboard timing performed.

Future Scope

Authentication, PostgreSQL, Docker deployment, cloud hosting, live market feeds, AI insights.

Conclusion

The platform provides a scalable foundation for financial screening and company analysis.